# BALAJI DILIPSINGH RAJPUT

**Biotechnology Graduate | Entry-Level QA/QC Trainee | Computational & Wet-Lab Analyst**

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## CAREER OBJECTIVE & PROFESSIONAL SUMMARY

Goal-oriented and detail-focused Biotechnology graduate with demonstrated academic excellence and practical foundational training in pharmaceutical manufacturing, analytical QC, and cGMP compliance. Recipient of the **Best Research Project Award (2024)** and **Dean's List** distinction. Practical exposure includes **2-Month Industrial Training in Oral Solid Dosage (Tablets) & QA/QC** at Elysium Pharmaceuticals Ltd., genomic data analysis workflows at Biotecnika, and GLP microbiology assays at Gujarat Biotech Innovations. Eager to contribute rigorous analytical skills and standard-operating-procedure discipline as an entry-level **Quality Assurance (QA) Trainee**, **IPQA Officer**, or **Quality Control (QC) Chemist**. Available to join immediately.

## CORE COMPETENCIES & FOUNDATIONAL KNOWLEDGE

- cGMP, GDP & GLP Principles
- Tablet Manufacturing & IPQA Checks
- Quality Control (QC) Wet Analysis
- Good Documentation Practices (ALCOA+)
- Dissolution, Friability & Hardness Testing
- SOP Drafting & Document Lifecycle
- Analytical Instrumentation (HPLC, UV-Vis)
- Microbial Testing & Aseptic Culture
- BMR / BPR Review Fundamentals
- Deviation & CAPA Problem-Solving
- Quality Management System (QMS)
- 21 CFR Part 11 & Schedule M Awareness

## EDUCATION

**Diploma in Biotechnology** — **2021 – 2025**

*Parul Institute of Technology & Engineering, Gujarat Technological University (GTU), Vadodara*

**Honors / Performance:** CGPA: 6.7 / 10 | Dean's List for Academic Excellence (2024–2025)

Key Subjects: Molecular Biology, Microbiology, Bioprocess Technology, Biochemistry, Immunology, Fermentation Technology, QC & QA.

## INDUSTRIAL TRAINING & INTERNSHIP EXPERIENCE

**Industrial Trainee — QA & QC (Oral Solid Dosage - Tablets) — Elysium Pharmaceuticals Ltd., Dabhasa, Vadodara** — **2 Months (Industrial Training)**

- Completed 2-month industrial training focused on Oral Solid Dosage (tablet manufacturing) and Quality Control (QC) & Quality Assurance (QA) workflows under cGMP and Schedule M guidelines.
- Gained practical QC laboratory exposure: observed raw material and finished product testing, dissolution and disintegration profiling, UV-Vis spectrophotometry assays, and GLP reagent standardization.
- Assisted IPQA officers in line clearance, physical in-process parameter checks (weight variation, thickness, hardness, friability), and BMR/BPR log verification.

**Bioinformatics Intern — Biotecnika (Remote)** — **2024 (3 Months)**

- Engineered automated Python / Biopython workflows for sequence alignment, screening over 500+ DNA and protein sequences.
- Utilized BLAST, ClustalW, and GenBank annotations to streamline curation workflows, cutting manual processing time by 40%.

**Laboratory Research Intern — Gujarat Biotech Innovations** — **2023 (2 Months)**

- Carried out routine laboratory assays: PCR, agarose gel electrophoresis, and microbial culture handling for 80+ samples/week.
- Authored 3 standard operating procedures (SOPs) compliant with GLP standards; elevated laboratory sample workflow efficiency by 20%.

## KEY ACADEMIC & RESEARCH PROJECTS

**Bioethanol Production from Agricultural Waste (2024)** **[Fermentation, DOE/RSM, HPLC, Microbial Kinetics, ANOVA]**

Optimized Simultaneous Saccharification & Fermentation (SSF) with *S. cerevisiae* via Response Surface Methodology (RSM/DOE); achieved a **15% yield enhancement** validated by HPLC and ANOVA ($p < 0.05$). **Awarded Best Research Project – Parul Institute 2024**.

**Comparative Genomic Analysis of AMR Genes (2024)** **[Python, Biopython, BLAST, GenBank, ClustalW, Bioinformatics]**

Investigated 120 bacterial genomes from NCBI GenBank using an automated Python/Biopython workflow; identified **8 novel Antimicrobial Resistance (AMR) gene variants** across 500+ sequence entries.

**Protein Structure Prediction & Molecular Docking (2023)** **[SWISS-MODEL, AutoDock Vina, PyMOL, PDB, In Silico Drug Design]**

Constructed 3D homology models for 5 therapeutic target proteins using SWISS-MODEL; performed virtual screening and molecular docking with 50 ligands via AutoDock Vina, identifying 3 lead inhibitors (binding affinity ≤ -8.0 kcal/mol).

**Biogenic Synthesis of Silver Nanoparticles (AgNPs) (2023)** **[Green Synthesis, UV-Vis Spectrophotometry, Microbial Assays, Aseptic Techniques]**

Synthesized antimicrobial AgNPs via green biogenic route using *Pseudomonas spp.*; confirmed nanoparticle formation and broad-spectrum bactericidal efficacy via UV-Vis Spectrophotometry and zone-of-inhibition assays.

## TECHNICAL & LABORATORY SKILLS

**Quality & Regulatory Awareness:** cGMP, GDP, GLP, Schedule M, SOP Drafting, BMR / BPR Verification, Deviation & CAPA Basics, Change Control Awareness, Data Integrity (ALCOA+), 21 CFR Part 11, CDSCO & WHO-GMP Guidelines.

**Laboratory & Analytical (QC/QA):** HPLC (Fundamentals), UV-Vis Spectrophotometry, Dissolution & Disintegration Testing, Friability, Hardness, PCR / RT-PCR, Agarose Gel Electrophoresis, Microbial & Aseptic Culture, Sterility Testing, Autoclave, LAF.

**Bioinformatics & Computation:** Python (Biopython, Pandas, NumPy), R Basics, BLAST, ClustalW, NCBI GenBank, UniProt, AutoDock Vina, SWISS-MODEL, PyMOL.

**Software & Digital Tools:** MS Office Suite (Advanced Excel, Word, PowerPoint), SPSS, MATLAB (Basic), Git & GitHub, TrackWise (Basics), Google Workspace.

## CERTIFICATIONS & SPECIALIZED TRAINING

- Bioinformatics Specialization – Coursera / UC San Diego (2024)
- Python for Genomic Data Science – Coursera / Johns Hopkins (2024)
- Industrial Biotechnology & GMP – NPTEL / IIT Madras (2023)
- Advanced Techniques in Biotechnology – Parul University (2025)
- Molecular Biology Techniques – Udemy (2023)
- Digital Marketing & Web Foundations – Google / ITI (2021–2024)

## KEY HONORS & ACHIEVEMENTS

- **Best Research Project Award (2024)** – Parul Institute of Technology & Engineering for Bioethanol SSF optimization.
- **Dean's List for Academic Excellence (2024–2025)** – Recognized for top percentile performance in Biotechnology coursework.
- **Automated Genomic Pipeline** – Developed Python workflow reducing manual sequence curation time by 40% (AMR study).

## PERSONAL DOSSIER & DECLARATION

**Languages Known:** English (Fluent), Hindi (Fluent), Gujarati (Native)
**Details:** Date of Birth: 20 June 2001 | Marital Status: Single | Nationality: Indian | Availability: Immediate Joiner

*I hereby solemnly declare that all statements and particulars provided in this resume are true, complete, and authentic to the best of my knowledge.*

**Place:** Vadodara, Gujarat
**Date:** _____

**(Balaji Dilipsingh Rajput)**
**Candidate Signature**